

VELOX PEPTIDES

The Velox Researcher's Handbook

A plain-English reference to research peptides,
handling and documentation

≥99% HPLC PURITY

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CoA PER BATCH

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About this handbook

This handbook is a laboratory reference. It explains, in plain English, what each research peptide Velox supplies is, what researchers have observed about it in preclinical (cell and animal) studies, and how these materials are handled and documented. Everything here describes observations from the research literature. Nothing in this document is medical, dosing or usage advice, and no health, cosmetic or performance benefit is claimed or implied. All materials are supplied for in vitro research use only to qualified researchers aged 21 or over.

PART ONE

Working with research peptides

A practical primer on what these materials are, how they are prepared and stored in the laboratory, and how their quality is documented. Written for reference only — not instructions for use in humans or animals.

1.1 What research peptides are

A **peptide** is a short chain of amino acids — the same building blocks that make up proteins. Where a protein may contain hundreds of amino acids, a peptide is usually just a handful, joined in a set order. That exact order (the *sequence*) is what gives each peptide its shape and decides which signals it can send inside a cell.

A **research peptide** is one made in a laboratory and supplied as a chemical reagent for scientific work. Researchers use them as precise, well-defined tools: because a peptide has a known sequence and high purity, a scientist studying a cell pathway knows exactly what they are adding to the dish. Many of the compounds in this handbook are lab-made copies of natural signalling molecules, often tweaked so they survive longer or act on a single pathway, which makes them cleaner to study.

Velox supplies every compound as a **lyophilised** (freeze-dried) powder. Before it can be used in a study it is **reconstituted** — mixed back into a liquid. The sections below cover how that is done, how the material is stored, and how its identity and purity are documented. None of this describes use in people or animals: these are bench reagents for in vitro research only.

1.2 Reconstitution basics

Reconstitution means turning the dried powder back into a liquid so it can be measured and used. The liquid researchers most often add is **bacteriostatic water** — sterile water that contains a tiny amount of benzyl alcohol (about 0.9%). That preservative stops bacteria from growing, so a reconstituted vial stays usable in the fridge for longer than it would with plain water.

The only number that matters when reconstituting is the **concentration** — how much peptide ends up in each millilitre of liquid. It follows one simple formula:

THE CONCENTRATION FORMULA

$$\text{mg in vial} \div \text{mL of water added} = \text{mg/mL}$$

The more water you add, the weaker (more dilute) each millilitre becomes; the less water, the more concentrated.

WORKED EXAMPLE

Suppose a vial holds **10 mg** of peptide and you add **2 mL** of bacteriostatic water. Using the formula: $10 \text{ mg} \div 2 \text{ mL} =$ **5 mg/mL**. Every 1 mL drawn from that vial now contains 5 mg of peptide; 0.2 mL would contain 1 mg.

Quick-reference concentration table (mg/mL)

VIAL SIZE	1 mL water	2 mL water	3 mL water	5 mL water
5 mg	5.0 mg/mL	2.5 mg/mL	1.7 mg/mL	1.0 mg/mL
10 mg	10.0 mg/mL	5.0 mg/mL	3.3 mg/mL	2.0 mg/mL
15 mg	15.0 mg/mL	7.5 mg/mL	5.0 mg/mL	3.0 mg/mL

20 mg	20.0 mg/mL	10.0 mg/mL	6.7 mg/mL	4.0 mg/mL
30 mg	30.0 mg/mL	15.0 mg/mL	10.0 mg/mL	6.0 mg/mL

Velox provides an online **reconstitution calculator** at veloxpeps.com/tools/reconstitution-calculator/ that does this maths for you — enter the vial size and the volume of water, and it returns the concentration per millilitre.

1.3 Storage & handling

How a peptide is stored decides how long it stays intact. The rules differ depending on whether it is still a dry powder or has been reconstituted into a liquid.

Lyophilised (dry) powder is the most stable state. Sealed in its vial, kept cold (ideally around -20°C), dry and away from light, a freeze-dried peptide can remain stable for a long time. Moisture and warmth are its main enemies, so vials are best kept in a frost-free freezer and allowed to reach room temperature before opening, which stops water condensing inside.

Reconstituted (liquid) peptide is less stable and should be kept refrigerated at $2-8^{\circ}\text{C}$ and used within a few weeks. The single most important rule is to avoid repeated **freeze-thaw** cycles: each time a solution freezes and thaws, ice crystals can damage the peptide chain, so a reconstituted vial is kept in the fridge rather than the freezer.

Good **sterile technique** protects both the researcher and the sample: work on a clean surface, wipe the vial stopper with alcohol before drawing from it, and use a fresh sterile needle each time. **Signs of degradation** to watch for include cloudiness, floating particles, a colour change, or any solid that will not dissolve — a clear, particle-free solution is the expected result for the compounds in this handbook.

Storage at a glance

STATE	TEMPERATURE	NOTES
Lyophilised powder	-20°C , dry, dark	Most stable state; warm to room temperature before opening to avoid condensation.
Reconstituted solution	$2-8^{\circ}\text{C}$ (fridge)	Use within a few weeks; avoid freeze-thaw cycles; keep out of direct light.
Short working period	Room temp, brief only	Minimise time at room temperature; return promptly to the fridge.

1.4 How to read a Certificate of Analysis

A **Certificate of Analysis** (CoA) is the lab report that proves what is actually in a vial. Every Velox batch ships with one, and a copy is available on request before ordering. It is the document that turns a label into evidence. There are four parts worth understanding.

HPLC purity %. HPLC (high-performance liquid chromatography) separates everything in the sample and measures how much of it is the target peptide versus anything else. A result such as $\geq 99\%$ means at least 99% of the material is the intended compound — the single most important quality number on the certificate.

Mass spectrometry (molecular weight). Mass spec weighs the molecule with great precision. If the measured molecular weight matches the expected weight for that exact amino-acid sequence, it confirms the vial contains the right peptide and not a look-alike — this is the identity check.

Appearance. A short physical description (for example, a white lyophilised powder) so the receiving researcher can confirm the material matches what the certificate describes.

What “PASS” means. A PASS records that the batch met every specification set for it — purity, identity and appearance all within limits. It is a documented result tied to that specific batch, not a general claim.

WHY THIRD-PARTY TESTING MATTERS

Velox sends samples to **Janoshik Analytical**, an independent laboratory, rather than relying only on in-house figures. Independent testing means the purity and identity results come from a lab with no stake in the outcome, which is what makes a CoA genuinely meaningful as a quality record.

CATEGORY 1 OF 5

Cognitive

Neuropeptides studied for brain-growth proteins, calming signals and sleep.

- Semax
- Selank
- DSIP

COGNITIVE SERIES

Semax

SPECIFICATIONS

CAS NUMBER	80714-61-0
PEPTIDE CLASS	ACTH(4-7) analogue
KEY PATHWAY	BDNF / NGF
HPLC PURITY	≥99.0% (batch-verified)

What it is

Semax is a lab-made peptide — a short chain of amino acids, the building blocks of proteins. It is made of seven of them (a heptapeptide, “hepta” meaning seven), with the sequence Met-Glu-His-Phe-Pro-Gly-Pro. It is a neuropeptide (a peptide that acts on nerve cells) and is copied from a small piece of a natural hormone called ACTH. In lab and animal studies, researchers have looked at how it relates to BDNF (brain-derived neurotrophic factor, a protein that helps brain cells grow and survive) and to protecting nerve cells. These are research observations only, not therapeutic effects. It is supplied strictly as a research reagent for in vitro (test-tube / lab work only) use, not for human or veterinary use.

Semax takes the active piece of ACTH (called the ACTH(4–7) fragment) and adds a Pro-Gly-Pro tail. That tail slows down how fast the body’s enzymes break it apart, so Semax lasts much longer than the plain fragment. Here is the useful part: this piece of ACTH still sends nerve-related signals, but without the part of the full hormone that triggers stress hormones. That makes it a clean research tool — scientists can study these nerve signals on their own, away from the body’s stress-hormone system.

Semax comes from a long line of Eastern European research and was studied there over many decades. In the lab it is used as a neuropeptide research tool, mainly to study how short peptides change brain-growth proteins, nerve-protecting gene activity, and learning in cell cultures and rodents.

What’s been studied

BDNF, NGF and neuroprotection

Semax research focuses on three linked areas: how it affects brain-growth proteins, how it behaves when nerve cells are damaged, and how it affects the brain chemicals tied to learning and attention.

Neurotrophic factor (BDNF/NGF) expression

The most studied thing about Semax is that it has been linked to more BDNF and NGF (nerve growth factor) in the brain. These are proteins that help nerve cells survive, grow, and form new connections (this connection-forming is called synaptic plasticity). In rodent studies, BDNF levels rose quickly in certain brain areas after Semax was given, along with changes in TrkB (the receptor that BDNF locks onto). Gene studies tracked how NGF and BDNF activity changed over time across different brain regions — and found the effect depends on the region and can go up or down, rather than being one simple uniform change.

Neuroprotection and ischaemia models

Semax has been tested in rodent models of stroke (where blood flow to part of the brain is cut off, called ischaemia) and oxidative stress (damage from harmful molecules). Researchers looked at how it relates to nerve cells surviving and to the activity of nerve-growth genes after injury. In these stroke models, Semax (and

a smaller piece it breaks down into, Pro-Gly-Pro) was linked to switching on nerve-growth genes and dialling down genes tied to inflammation and cell death. This is why it is used as a research tool for studying how nerve cells can be protected. These are research observations only, not therapeutic effects.

Cognition, attention and monoamines

Beyond brain-growth proteins, studies have looked at Semax and attention, learning, and the dopamine and serotonin systems (two chemical messengers involved in mood, focus and reward). It has also been studied for how it affects enkephalins (natural pain-relieving signals). All together, this makes it a useful research tool for studying how one short peptide can affect thinking through several pathways at once.

Velox supply. Supplied by Velox Peptides as a lyophilised powder, batch-verified by HPLC and independently tested by Janoshik, with a Certificate of Analysis available per batch. For in vitro research use only.

COGNITIVE SERIES

Selank

SPECIFICATIONS

CAS NUMBER	129954-34-3
PEPTIDE CLASS	Tuftsins analogue
KEY PATHWAY	GABAergic / BDNF
HPLC PURITY	≥98.9% (batch-verified)

What it is

Selank is a lab-made peptide — a short chain of amino acids, the building blocks of proteins. It is made of seven of them (so scientists call it a heptapeptide, “hepta” meaning seven), with the sequence Thr-Lys-Pro-Arg-Pro-Gly-Pro. It is a neuropeptide (a peptide that acts on nerve cells) and is copied from a natural immune-system peptide called tuftsins. In animal studies, researchers have looked at how it calms anxiety-like behaviour (an anxiolytic effect) while affecting the GABA system, which is a calming signal in the brain. These are research observations only, not therapeutic effects. It is supplied strictly as a research reagent for in vitro (test-tube / lab work only) use, not for human or veterinary use.

Selank was made in Russia at the Institute of Molecular Genetics. Scientists took the natural peptide tuftsins (Thr-Lys-Pro-Arg) and added a short Pro-Gly-Pro tail. That tail makes the peptide last much longer before the body's enzymes break it down. Because it comes from tuftsins, researchers study it not just for behaviour but also for how it affects the immune system and brain chemistry.

Like Semax, the peptide it is often paired with, Selank comes from a long line of Eastern European research. Scientists use it as a tool to study how short peptides change anxiety-like behaviour, brain chemical messengers, and stress responses in animals.

What's been studied

Selank research focuses on three main areas: how it affects the calming GABA system, how it affects brain growth signals and mood chemicals, and how (because it comes from tuftsins) it affects pain-related signals and the immune system.

GABAergic modulation

The most studied thing about Selank is how it affects the GABA system. GABA is the brain's main “calming” chemical messenger, and it is the usual target for anti-anxiety compounds. Here is the key point: Selank does not lock onto the same spot that benzodiazepines (common sedative drugs) use. Instead, studies describe it changing how the GABA-A receptor is built and how strongly it responds to GABA — an indirect (allosteric) way of nudging the receptor. Researchers think this is why animal studies show a calming effect without the heavy sedation and clumsy movement you get with benzodiazepines. These are research observations only, not therapeutic effects.

BDNF and monoamine signalling

In rodent studies, Selank has also been linked to more BDNF (brain-derived neurotrophic factor, a protein that helps brain cells grow and survive) in the hippocampus, a part of the brain tied to memory. It has also been linked to serotonin signalling (serotonin is a chemical messenger involved in mood). These pathways relate to memory, mood and coping with stress. This mix of calming and brain-growth effects sets Selank apart from

compounds that only sedate.

Enkephalin stability and immune signalling

Because it comes from tuftsin, Selank has also been studied for how it affects enkephalins (natural pain-relieving signals in the body) — it is reported to slow their breakdown — and for effects on the immune system and cytokines (chemical messengers the immune system uses). In short, this is a peptide that links the nervous system and the immune system in research models.

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COGNITIVE SERIES

DSIP

SPECIFICATIONS

CAS NUMBER	62568-57-4
PEPTIDE LENGTH	9 amino acids
ORIGIN	Endogenous neuropeptide
HPLC PURITY	≥98.7% (batch-verified)

What it is

DSIP (short for Delta Sleep-Inducing Peptide) is a peptide — a short chain of amino acids, the building blocks that make up proteins. It is made of 9 of these building blocks (so scientists call it a nonapeptide; "nona" just means nine), and its order is Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu. In animal studies, researchers have looked at how it relates to deep sleep and to how the body handles stress. It is sold only as a research reagent (a chemical for lab tests), for in vitro use (test-tube / lab work only) — never for use in humans or animals.

Scientists first found DSIP in the blood of rabbits during sleep experiments. Its name comes from "delta waves" — the slow brain waves your brain makes during deep sleep. DSIP is an endogenous peptide, which just means the body makes it naturally. That makes it a useful tool for studying how small peptides take part in sleep and in the body's hormone signals.

Even after many years of study, no one is fully sure exactly how DSIP works or which receptor (the "lock" a molecule fits into to send a signal) it acts on. One review even called it a "still unresolved riddle." That unanswered question is part of why scientists keep studying it.

What's been studied

Sleep modulation and stress-response research

DSIP research covers three main areas: how it relates to deep sleep, how it links to sleep-time hormone signals, and a wider role in stress and keeping the body balanced. All of this is studied even though scientists still don't fully understand how it works.

Slow-wave (delta) sleep modulation

The thing DSIP is best known for in research is its link to slow-wave (delta) sleep — the deepest stage of sleep. In rat and rabbit studies, DSIP and stronger lab-made copies of it were linked to more deep sleep and more REM sleep (the dream stage). When scientists blocked the animal's own DSIP using an antiserum (a substance that grabs and neutralises a specific molecule), the usual rebound of extra deep sleep after sleep loss was reduced. This suggests DSIP takes part in the body's natural control of deep sleep, rather than just knocking the animal out.

Sleep-related growth hormone and neuroendocrine signalling

DSIP has also been studied for its links to hormones that get released during sleep, including growth hormone (GH), which helps the body grow and repair. Research has described DSIP affecting a signal called somatostatin through a dopamine-based pathway — one way a sleep-linked peptide could connect to the growth-hormone system. This makes it a handy tool for studying how sleep and hormone release are tied together.

Stress, antioxidant and homeostatic effects

Apart from sleep, DSIP has been looked at for how it relates to calming the body's stress response, to antioxidant activity (mopping up cell-damaging particles), and to keeping the body's internal systems in balance, all in animal studies. Together, these make DSIP a research reagent useful for studying how a peptide can affect several of the body's natural rhythms.

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CATEGORY 2 OF 5

Metabolic

Compounds studied for how cells and the body handle energy, sugar and fat.

- Retatrutide
- MOTS-c
- NAD+

METABOLIC SERIES

Retatrutide

SPECIFICATIONS

ALSO KNOWN	LY3437943 (Eli Lilly)
PEPTIDE CLASS	Triple agonist
TARGETS	GLP-1, GIP & glucagon receptors
HPLC PURITY	≥99% (batch-verified)

What it is

A peptide is a short chain of amino acids — the same building blocks that make up proteins. Retatrutide (also called LY3437943, made by Eli Lilly) is a lab-made peptide built to act on three different "receptors" at the same time. A receptor is like a lock on the surface of a cell, and a molecule that fits the lock and switches it on is called an agonist (think of it as a key that turns the lock on). Retatrutide acts on three of these locks: the GLP-1 receptor, the GIP receptor, and the glucagon receptor.

That is why it is called a triple agonist: it switches on (agonises) all three of these locks at once, not just one or two. Each of the three receptors helps control metabolism — the way the body handles energy, blood sugar (glucose), and fat. Because it hits all three at the same time, Retatrutide works in a more complicated way than older compounds that only act on one or two receptors.

Retatrutide has been studied more than most research peptides like it. It has been through Phase 1 and Phase 2 clinical trials (early human studies done in steps to check safety and effects), and the results were published in well-known journals: the New England Journal of Medicine, The Lancet, and Cell Metabolism. All this published data makes it one of the most studied peptides you can find for incretin & receptor research.

What's been studied

To understand how Retatrutide works, it helps to look at each of its three receptor "locks" on its own first, then see how they work together. The summaries below come from published research studies.

GLP-1 Receptor Pathway

GLP-1 (Glucagon-Like Peptide-1) receptors are found all over the body — in the insulin-making cells of the pancreas (beta cells), the brain and nerves, the gut, and other tissues. In animal studies, switching on GLP-1 receptors is linked to releasing more insulin (the hormone that lowers blood sugar), but only when blood sugar is up (this is what "glucose-dependent" means). It is also linked to the stomach emptying more slowly and to appetite-lowering signals in a part of the brain called the hypothalamus.

The GLP-1 receptor pathway has been studied a lot in rodents and in monkeys, and switching on the GLP-1 receptor is the main way that the most-researched metabolic peptides work. Retatrutide uses this as one of its three actions.

GIP Receptor Pathway

GIP (Glucose-Dependent Insulinotropic Polypeptide) receptors are found in the pancreas beta cells, in fat tissue, in bone, and in the brain. In lab studies, switching on the GIP receptor is linked to releasing more insulin and, importantly, to changes in how fat tissue works — including signals for storing fat and for breaking it down (lipolysis means breaking down fat) in mice.

Scientists got interested in switching on GIP and GLP-1 together because the two seem to help each other rather than just doing the same job twice. Finan et al. showed in mice, monkeys, and early human studies that a single molecule switching on both GIP and GLP-1 gave bigger results in studies than either one alone. Retatrutide builds on this by adding the glucagon receptor as a third action.

Glucagon Receptor Pathway

Glucagon receptors sit mostly in the liver cells (hepatocytes), where they control how much stored sugar the liver releases into the blood. In animal studies, switching on the glucagon receptor also seems to change how much energy the body burns, with some data pointing to more heat being made (thermogenesis means making body heat).

On its own, switching on the glucagon receptor raises blood sugar — the opposite of what the GLP-1 receptor does. So why put it in a triple agonist? The idea researchers are testing is that, when it works together with the GLP-1 and GIP actions, the overall effect on energy-burning and fat in studies may be bigger than GLP-1/GIP alone, while the insulin released by the GLP-1 and GIP pathways helps keep blood sugar in check at the same time. Testing this idea was a big reason Retatrutide was made and studied.

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METABOLIC SERIES

MOTS-c

SPECIFICATIONS

PEPTIDE LENGTH	16 amino acids
ORIGIN	Mitochondrial 12S rRNA
KEY PATHWAY	AMPK signalling
HPLC PURITY	≥99% (batch-verified)

What it is

MOTS-c is a small peptide (a short chain of amino acids, the building blocks of proteins) made of 16 amino acids. What is special is where it comes from: it is built using instructions stored inside the mitochondria (the tiny "power plants" inside cells that make energy). In lab studies, scientists look at MOTS-c as a chemical that helps control how cells handle energy and fuel, mostly by switching on a sensor called AMPK (explained below). It is sold strictly as a research chemical for in vitro use (test-tube and lab work only) — never for people or animals.

Here is why MOTS-c is unusual. Nearly every peptide in the body is built from instructions kept in the cell's main control centre, the nucleus. MOTS-c is different — its instructions live inside the mitochondria instead. It is part of a small, fairly new group called mitochondrial-derived peptides (MDPs). These act like tiny messengers, carrying news from the mitochondria to the rest of the cell, including back to the nucleus.

MOTS-c was first described by Lee, Cohen and their team in 2015. It was one of the first of these mitochondrial messengers shown to have a clear job in controlling energy and fuel. That is why researchers are so interested in it as a way to study how the cell's power plants report on their energy status.

What's been studied

MOTS-c research focuses on one main thing — switching on the AMPK energy sensor — plus an eye-catching way it sends signals from the mitochondria to the nucleus, and several energy-related results seen in animal studies.

AMPK activation via the folate cycle

The best-studied thing about MOTS-c is that it switches on AMPK. AMPK is the cell's "low-fuel light" — a sensor that turns on when a cell is running low on energy and tells it to bring in more sugar and burn more fat. The way MOTS-c does this is clever and roundabout: it gets in the way of a chemical process called the folate cycle. This causes a helper molecule named AICAR to build up, and that build-up is what flips AMPK on. In these studies, skeletal muscle (the muscle you move with) seems to be the main place this happens.

Mitochondrial-to-nuclear signalling

When a cell is under stress (for example, running short on energy), MOTS-c has been seen to move out of the mitochondria and travel into the nucleus — the cell's control centre. There it teams up with proteins that switch genes on or off, helping turn on the cell's defence and antioxidant genes (which protect against damage). This makes MOTS-c a rare, clear example of a peptide that physically carries a message from the power plants to the control centre — a big reason it is studied.

Insulin sensitivity, exercise and ageing

In whole-animal rodent studies, MOTS-c was linked to keeping the body better at responding to insulin (the hormone that helps cells take in sugar), even with age or a high-fat diet, and to less weight gain from that diet. It has also been described as a peptide that responds to exercise and is linked to fitness and age-related decline. This makes it a useful research probe for studying energy control and ageing. These are research observations only, not therapeutic effects in people.

Velox supply. Supplied by Velox Peptides as a lyophilised powder, batch-verified by HPLC and independently tested by Janoshik, with a Certificate of Analysis available per batch. For in vitro research use only.

METABOLIC SERIES

NAD⁺

SPECIFICATIONS

CAS NUMBER	53-84-9
CLASS	Pyridine dinucleotide coenzyme
KEY PATHWAY	Sirtuins / cellular redox
HPLC PURITY	≥99.5% (batch-verified)

What it is

NAD⁺ (short for nicotinamide adenine dinucleotide) is a helper molecule that cells use to make energy. Scientists call this kind of helper molecule a coenzyme — a small molecule an enzyme needs in order to do its job. NAD⁺ is found in every living cell, and it has two main jobs. First, it helps move energy around inside the cell. Second, it acts as fuel for special enzymes (proteins that speed up chemical reactions) called sirtuins and PARPs. NAD⁺ is supplied strictly as a research reagent for in vitro use (test-tube and lab work only), not for human or veterinary use.

For its first job, NAD⁺ comes in two forms that the cell keeps swapping back and forth: a "loaded" form (NAD⁺) and an "unloaded" form (NADH). Think of NAD⁺ as a tiny battery that picks up bits of energy (electrons) from food and carries them to where the cell turns them into ATP, the cell's main energy currency. This swapping happens over and over, so NAD⁺ is one of the busiest molecules in the cell.

But the reason NAD⁺ gets so much research attention is its second job: certain enzymes use it up as fuel to control which genes are switched on and to repair damaged DNA. Having both jobs — energy carrier and fuel for control enzymes — is what ties a cell's energy level to research on ageing, metabolism (how the body turns food into energy), and how cells handle stress. That is why scientists study NAD⁺ and its precursors (the smaller molecules cells turn into NAD⁺).

What's been studied

Sirtuins, cellular energy and the precursor rationale

Research on NAD⁺ follows four connected threads: how it moves energy around, how it fuels sirtuins, how DNA-repair enzymes use it up, and how its levels drop with age (the main reason scientists study its precursors).

Redox metabolism and the NAD⁺/NADH cycle

When a cell breaks down food for energy — through steps with names like glycolysis, the citric-acid cycle, and oxidative phosphorylation — NAD⁺ grabs energy bits (electrons) to become NADH, then drops them off and turns back into NAD⁺. "Redox" just means these reactions that pass electrons back and forth. The balance between how much NAD⁺ and how much NADH a cell has tells scientists a lot about the cell's energy state, so they use it in research as a quick read on energy balance.

Sirtuin signalling

Sirtuins are a family of enzymes that need NAD⁺ to work. They help link the way proteins are tagged with chemical labels to metabolism, stress responses, and ageing. Every time a sirtuin does its job, it uses up a bit of NAD⁺. So the amount of NAD⁺ a cell has directly limits how active its sirtuins can be — which is why, in animal studies, the size of a cell's NAD⁺ supply is studied as a control knob for sirtuin-driven activity.

PARP enzymes and DNA repair

PARP enzymes also run on NAD⁺. They switch on when DNA is damaged and burn through a lot of NAD⁺ while making repairs. This puts NAD⁺ right in the middle of two jobs — keeping DNA healthy and making energy. It also means that when DNA gets damaged, the repair work can use up the NAD⁺ that sirtuins would have used. Researchers study this tug-of-war when looking at how cells handle stress.

Age-related decline and the precursor rationale

In lab animals, the amount of NAD⁺ in cells is reported to drop as they get older. Scientists think this drop may help explain why energy-making and the cell's tiny power plants (mitochondria) work less well with age. So a lot of research looks at NAD⁺ precursors — smaller molecules like NMN and NR that the cell turns into NAD⁺ — to see whether topping NAD⁺ back up changes sirtuin activity and mitochondrial function. NAD⁺ itself is used as the reference reagent (the standard to compare against) in this work.

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CATEGORY 3 OF 5

Recovery

Peptides studied in tissue-repair, blood-vessel and inflammation models.

- BPC-157
- TB-500
- KPV
- GHK-Cu

RECOVERY SERIES

BPC-157

SPECIFICATIONS

CAS NUMBER	137525-51-0
SEQUENCE	GEPPPGKPADDAGLV (15 aa)
CLASS	Synthetic pentadecapeptide
HPLC PURITY	≥99.4% (batch-verified)

What it is

BPC-157 (short for Body Protection Compound-157) is a peptide — a short chain of amino acids, the same building blocks that make up proteins. This one is made of 15 of those building blocks (its sequence is GEPPPGKPADDAGLV), which is why scientists call it a pentadecapeptide ("penta-deca" just means fifteen). It is a lab-made copy of a small piece of a natural protein found in stomach juice. In animal studies, researchers have looked at how it relates to tissue repair (the body fixing damage), angiogenesis (the growing of new blood vessels), and protecting the gut. Velox supplies it strictly as a research chemical for in vitro use — that means test-tube and lab work only, never for people or animals.

One thing that makes BPC-157 stand out in research is that it is tough. It stays in one piece when mixed into water or exposed to stomach-like conditions in the lab, while many other peptides fall apart. That stability is a big reason scientists use it so often in studies of how cells heal and move.

What's been studied

In lab and animal studies, the same few biological pathways (the chains of chemical signals cells use to talk to each other) keep coming up with BPC-157. Here is a plain summary of what researchers have observed — these are research findings only, not proof it treats anything.

Angiogenesis & the VEGFR2–eNOS–NO pathway

Studies in cells and in rodents have looked at BPC-157 and angiogenesis — the growing of new blood vessels. They noted links to a signal called VEGFR2 (think of it as a "docking point" on a cell that tells it to start building new vessels) and to nitric oxide, a molecule that helps blood vessels relax and form. These were seen in endothelial cells, the cells that line the inside of blood vessels.

Tendon & ligament fibroblasts

Lab work on tendon fibroblasts (the cells that build and repair tendons) has reported changes in how those cells grow, survive and move, plus changes in their growth-hormone receptors. These are common things scientists measure when studying how tissue repairs itself.

Gastrointestinal mucosal integrity

Because BPC-157 comes from a piece of a stomach protein, a lot of the early research looked at the gut lining (the protective layer inside the digestive system) and the gut-brain axis (the two-way connection between the gut and the brain), using animal injury models.

BPC-157 is often studied alongside TB-500 in tissue-repair models, where the two are thought to work in different but complementary ways — one linked to new blood vessels, the other to how cells move. For a side-by-side breakdown, see BPC-157 vs TB-500.

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RECOVERY SERIES

TB-500

SPECIFICATIONS

CAS NUMBER	77591-33-4
PEPTIDE CLASS	Actin-binding peptide
PARENT PROTEIN	Thymosin β 4 (43-aa)
HPLC PURITY	$\geq 99\%$ (batch-verified)

What it is

TB-500 is a lab-made peptide — a short chain of amino acids, the building blocks of proteins. It is based on a natural protein called Thymosin β 4. In lab and animal studies, researchers have looked at how it relates to cell migration (cells moving from one place to another), angiogenesis (the growing of new blood vessels), and tissue repair (the body fixing damage). It is supplied strictly as a research reagent for in vitro (test-tube / lab work only) use, not for human or veterinary use.

Thymosin β 4 is a natural protein made of 43 amino acids. It is found in almost every cell in mammals, and there is a lot of it in platelets (the cells that help blood clot) and in the fluid around a wound — the first things to show up when there is an injury. The name “TB-500” refers to a lab-made version built around the active part of Thymosin β 4. Researchers use it as a clean, well-defined tool for studying how cells move and how tissue is rebuilt.

Here is the key idea: actin (a protein that forms the “scaffolding” cells use to move and hold their shape) comes in two forms — loose single pieces (G-actin) and long chains (F-actin). Thymosin β 4 grabs the loose pieces and keeps them in reserve, helping control the balance between the two. Almost every cell that moves — cells building new blood vessels, skin cells closing a wound, repair cells travelling to damaged tissue — has to quickly rebuild its actin scaffolding to do so. That one job is why Thymosin β 4 sits at the centre of so much tissue-repair research. These are research observations only, not therapeutic effects.

What’s been studied

TB-500 / Thymosin β 4 research falls into four main areas, all coming from its actin job. The parts below explain how each is studied in cell cultures and animals. All of this is research observations only, not therapeutic effects.

Actin sequestration and the cytoskeleton

By holding onto loose actin pieces, Thymosin β 4 keeps a ready supply that a cell can use to quickly build its “feet” — the front-edge parts (called lamellipodia and filopodia) that pull a cell forward. In wound-healing studies, having this supply on hand is linked to skin cells and blood-vessel cells moving into the wound faster. This is the root of nearly every other effect the peptide is known for.

Cell migration and angiogenesis

Studies have also looked at how Thymosin β 4 relates to blood-vessel cells moving, forming tiny tubes, and growing brand-new blood vessels (angiogenesis). New vessels matter because repaired tissue needs a fresh blood supply to survive. In rodent studies, the peptide was reported to grow more new vessels and speed up skin-wound repair in both young and old animals. That helped make it a tool for blood-vessel research, not just a basic scaffolding protein.

Inflammation and extracellular-matrix remodelling

Thymosin β 4 has been studied for helping inflammation settle down — lowering the output of cytokines and chemokines (chemical messengers that drive inflammation) — and for affecting enzymes (called matrix metalloproteinases) that rebuild the “mesh” of material between cells during healing. Scientists are still mapping exactly which receptors carry out these effects, so this is still an active area of research.

Cardiac, corneal and other tissue models

The same “help cells move and survive” theme has been studied in tissues beyond skin. Heart studies looked at how Thymosin β 4 relates to heart-muscle cells surviving and moving after the blood supply is cut off, and eye studies looked at healing on the cornea (the clear front of the eye). Together these show why one actin-binding peptide draws research interest across so many kinds of repair.

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RECOVERY SERIES

KPV

SPECIFICATIONS

CAS NUMBER	13147-97-2
SEQUENCE	Lys-Pro-Val
ORIGIN	C-terminal α -MSH fragment
HPLC PURITY	$\geq 98.5\%$ (batch-verified)

What it is

KPV is a tiny peptide (a short chain of amino acids, the building blocks of proteins). It is made of just three of them — lysine, proline and valine — which is why it is called a tripeptide. It is the tail-end piece of a larger hormone called alpha-melanocyte-stimulating hormone (α -MSH for short). Scientists study KPV because it appears to calm inflammation (the body's swelling-and-redness reaction) in lab tests. It is sold only as a research chemical for in vitro work (test-tube and lab use only) — never for people or animals.

The full α -MSH hormone does two jobs at once: it calms inflammation, and it flips a cell switch called the melanocortin receptor that controls skin colour. KPV is like the calming half of that hormone on its own. Because it is only three amino acids long, it keeps much of the anti-inflammatory effect but drops the colour-changing part. That makes it a clean tool for researchers who only want to study inflammation.

One important lab finding makes this clearer: KPV calms inflammation without using the melanocortin receptor at all. Instead it gets into cells a different way (explained below). This is one reason KPV is studied as its own well-defined research chemical rather than just a leftover piece of α -MSH.

What's been studied

NF- κ B inhibition and mucosal inflammation models

KPV research looks at three things: the inflammation "alarm signals" it turns down, the unusual doorway it uses to get into cells, and the gut and skin lab models where it has been studied most.

NF- κ B and MAP-kinase inhibition

The best-studied thing about KPV is that it turns down a signal called NF- κ B. Think of NF- κ B as a master "on switch" inside cells that tells them to start an inflammation response. KPV also quiets a second alarm pathway called MAP-kinase. In lab tests on cells, even very small amounts of KPV lowered NF- κ B activity and reduced the amount of cytokines the cells released. (Cytokines are chemical messengers cells use to call in inflammation.) This is why researchers use KPV as an anti-inflammatory test tool.

PepT1-mediated uptake (melanocortin-receptor-independent)

A key finding is how KPV gets inside cells. It rides in through a doorway called PepT1 — a transporter (a tiny gate that carries small peptides into a cell) found on gut-lining cells and immune cells. And it does its calming job without touching the melanocortin receptor (the skin-colour switch). Using this special doorway, and skipping that switch, is what sets KPV apart from the full α -MSH hormone. It also explains why KPV still works when it reaches the inside of the gut.

Intestinal and colitis models

KPV has been studied most in models of gut and lining inflammation. In mice given a chemical to trigger colitis (inflammation of the colon, part of the gut), feeding KPV by mouth was reported to make the colitis less severe and lower the levels of inflammation messengers. That makes it a commonly cited tool in gut-inflammation research.

Skin and tissue-repair context

Because KPV comes from the α -MSH hormone, it is also studied in skin and wound inflammation. There, scientists check the same calming-down effect in skin cells, often next to other repair peptides — as research observations only, not as a treatment for people.

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RECOVERY SERIES

GHK-Cu

SPECIFICATIONS

CAS NUMBER	49557-75-7
SEQUENCE	Gly-His-Lys · Cu(II)
CLASS	Copper-binding tripeptide
HPLC PURITY	≥99% (batch-verified)

What it is

GHK-Cu (also called copper-tripeptide-1) is a peptide — a short chain of amino acids, the building blocks that make up proteins. This one is made of just 3 of those building blocks (glycine, histidine and lysine), so it is called a tripeptide ("tri" means three). It is joined to a single copper ion, which is what the "Cu" stands for (Cu is the chemical symbol for copper). The "Cu" copper part is why scientists call it GHK-Cu. It was first found in human blood plasma (the liquid part of blood) in the 1970s by two researchers, Pickart and Thaler. In lab dish tests, they saw it could change how liver cells from older donors behaved.

The amount of GHK in our blood drops as we get older. That is one reason scientists keep studying it as a research tool in models of tissue repair (the body fixing damage), skin-cell biology, and antioxidant pathways (the systems that mop up cell-damaging particles). The lab-made version used for research is chemically the same as the natural one, with the copper part still attached.

What's been studied

In lab-dish (in vitro) and animal studies, GHK-Cu has been linked to several body pathways — the chains of signals cells use to do their jobs. The points below are things researchers have observed in studies only, not effects in people.

Collagen & elastin gene expression

In studies on skin cells called dermal fibroblasts (the cells that build the support structure of skin), GHK-Cu was linked to switching on genes for type I collagen, elastin, decorin and glycosaminoglycans. These are the materials that give skin and other connective tissue its strength and stretchiness. (A gene being "expressed" just means the cell is using its instructions to make that material.)

Antioxidant & copper-transport activity

GHK-Cu is studied as a tool for looking at how copper moves around inside tissue, and for how it relates to antioxidant enzymes (proteins like superoxide dismutase that help mop up cell-damaging particles) in lab tests where cells are put under oxidative stress.

Wound-response & angiogenesis models

In rodent and lab-dish studies that mimic how tissue heals, GHK-Cu has been observed alongside angiogenesis (the growing of new blood vessels), the movement of skin-building fibroblast cells, and the rebuilding of the tissue's support structure — all common things scientists measure in repair research.

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CATEGORY 4 OF 5

Growth

GHRH-type peptides studied for how the pituitary releases growth hormone.

- CJC-1295
- Tesamorelin

GROWTH SERIES

CJC-1295

SPECIFICATIONS

SEQUENCE	Modified GHRH(1-29)
PEPTIDE CLASS	GHRH analogue
TARGET	GHRH receptor
HPLC PURITY	≥99.1% (batch-verified)

What it is

CJC-1295 without DAC — also called Modified GRF(1-29) — is a lab-made peptide (a short chain of amino acids, the building blocks of proteins). In studies, it acts like GHRH, a natural hormone that tells the body to release its own growth hormone in short bursts. It is sold only as a research chemical for in vitro (lab) use, not for use in people or animals.

GHRH stands for growth-hormone-releasing hormone. Its first 29 amino acids hold almost all of its power to switch on its target, so scientists call that piece GHRH(1-29). CJC-1295 starts from that piece and swaps in four amino acids that protect it from enzymes — tiny molecular “scissors” in the blood (such as one called DPP-IV) that would normally chop up natural GHRH within minutes. The result is a peptide that lasts noticeably longer than plain GHRH(1-29) but still works in short bursts.

The words “without DAC” are the key point. DAC (short for Drug Affinity Complex) is an extra chemical handle that grabs onto albumin (a common blood protein) so the peptide stays active for days. The version without DAC leaves that handle off, so it works for only about half an hour instead of days. That short, bursty action is what makes it useful for studying how the GHRH target switches on. These are research observations only, not therapeutic effects.

What’s been studied

■ GHRH-receptor activation and GH pulse research

CJC-1295 research follows the same plan as other GHRH-like peptides: switch on the target, see how long the effect lasts, and compare the DAC and no-DAC versions on purpose.

■ Receptor mechanism

Like Tesamorelin, CJC-1295 without DAC docks onto the GHRH receptor (the matching “lock” for this peptide) on somatotroph cells in the pituitary — a small gland at the base of the brain. Somatotroph cells are the ones that make growth hormone. Docking there raises a cell messenger called cAMP and gets those cells to make and release the body’s own growth hormone. It works one step before growth hormone rather than adding growth hormone directly, so the gland keeps its own natural controls in research models.

■ Half-life and pulse amplitude

The main research feature of the no-DAC form is that it acts for only a short time — about 30 minutes. (“Half-life” just means how long it takes for half of it to break down or clear.) That short window is studied for producing separate bursts of growth hormone that look a lot like the body’s natural rhythm. Studies report that this natural burst pattern is kept under CJC-1295, with higher growth hormone and higher IGF-1 (a protein the body makes in response to growth hormone), rather than the rhythm being flattened out. These are research observations only, not therapeutic effects.

DAC vs without-DAC: complementary research tools

The DAC form keeps the target switched on for days, while the no-DAC form gives one short burst. That makes them a useful pair in the lab: researchers can separate the pattern of switching on the target (steady vs bursty) from the simple fact that it is switched on, and then study how that pattern changes growth hormone and IGF-1 over time.

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GROWTH SERIES

Tesamorelin

SPECIFICATIONS

CAS NUMBER	218949-48-5
PEPTIDE CLASS	GHRH analogue
TARGET	GHRH receptor
HPLC PURITY	≥99% (batch-verified)

What it is

Tesamorelin is a GHRH analogue — a lab-made copy of the natural signal (called GHRH) that tells the body to release growth hormone. It is a peptide (a short chain of amino acids, the building blocks of proteins). It fits a lock called the GHRH receptor on the pituitary (a small gland at the base of the brain). Researchers study it as a tool for nudging the body to release its own growth hormone in natural bursts. It is supplied strictly as a research reagent for in vitro (test-tube / lab work only) use, not for human or veterinary use.

Chemically, Tesamorelin is the full natural GHRH chain (44 amino acids long) with one extra piece added to one end (called a trans-3-hexenoyl group). That small change is what makes it useful. Normal GHRH gets chopped up within minutes by enzymes in the body. The added cap acts like armour on the weak end of the peptide, so Tesamorelin lasts much longer than plain GHRH in research.

Tesamorelin works one step before growth hormone. Instead of being growth hormone itself, it tells the pituitary to make and release its own. Researchers use it to study that earlier step — how the pituitary responds to the GHRH signal — which is what makes it different from simply giving growth hormone directly. These are research observations only, not therapeutic effects.

What's been studied

GHRH-receptor signalling vs direct GH

The way Tesamorelin works follows a clear chain: it fits its receptor, that sets off a signal inside the cell, the pituitary releases growth hormone in bursts, and that growth hormone then triggers a second signal called IGF-1. The effects on metabolism (how the body uses energy and handles fat) are the most studied part.

Receptor binding and the cAMP cascade

Tesamorelin fits the GHRH receptor on cells in the pituitary called somatotrophs (the cells that make growth hormone). When the receptor switches on, it raises a tiny “messenger” molecule inside the cell called cAMP. That rise in cAMP makes the cell both build and release growth hormone. Because it works through the cell's own machinery, researchers use Tesamorelin to test how well these cells respond and whether the GHRH signalling is working properly.

Pulsatile release vs direct growth hormone

The key point is that Tesamorelin works one step earlier. It asks the pituitary to release the body's own growth hormone in its natural rhythm of bursts, keeping the body's normal feedback controls in place. Giving growth hormone directly skips the pituitary completely and gives a flat, steady level instead of natural bursts. For researchers studying how the growth-hormone system is controlled, keeping those natural bursts is exactly what makes a GHRH analogue a truer tool than plain growth hormone.

Downstream GH and the IGF-1 axis

The growth hormone that Tesamorelin helps release then acts on the body's tissues and triggers IGF-1 (insulin-like growth factor 1), the messenger behind many of growth hormone's effects. So research with Tesamorelin reaches past the pituitary to the wider growth-hormone/IGF-1 system and how it affects metabolism.

Metabolic research context

The most studied result of GHRH-analogue signalling is on metabolism — especially effects on visceral fat (the fat stored deep around the organs) and how the body handles fats. This is the area Tesamorelin has been studied most. In research settings it is used as a tool to look at how prompting the body's own growth hormone changes where fat sits and other metabolic markers. These are research observations only, not therapeutic effects.

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CATEGORY 5 OF 5

Antioxidant

Molecules studied for how cells defend against oxidative damage.

— Glutathione

— Melanotan II

ANTIOXIDANT SERIES

Glutathione

SPECIFICATIONS

CAS NUMBER	70-18-8
SEQUENCE	γ -Glu-Cys-Gly
KEY PATHWAY	Cellular redox / antioxidant
HPLC PURITY	$\geq 99.2\%$ (batch-verified)

What it is

Glutathione (GSH) is a peptide — a short chain of amino acids, the building blocks that make up proteins. It is made of 3 of them (glutamate, cysteine and glycine), so it is a tripeptide ("tri" means three). It is the most common antioxidant inside our cells. An antioxidant is a molecule that mops up cell-damaging particles. Scientists study it as a key part of how cells stay balanced and as the helper that lets a group of cleaning-up enzymes do their work. It is sold only as a research reagent (a chemical for lab tests) for in vitro use (test-tube / lab work only), never for use in humans or animals.

Its full chemical name is L- γ -glutamyl-L-cysteinyl-glycine. The part that does the work is a sulphur-containing piece (called a thiol, written –SH) on its cysteine building block. That piece lets glutathione hand out electrons to neutralise reactive oxygen species (unstable particles that can damage cells). When it does this, it switches between its "fresh" form (GSH) and its "used-up" form (GSSG).

Glutathione is found in nearly every cell and is often called the cell's "master antioxidant." It is so important that scientists measure the ratio of fresh (GSH) to used-up (GSSG) glutathione as a quick way to tell how much stress a cell is under. This makes glutathione both something worth studying on its own and a reliable reference chemical for oxidative-stress studies (tests on cell damage from unstable particles).

What's been studied

Redox cycling and oxidative-stress research

Glutathione research follows its chemistry: a recycling loop, two enzyme systems that need it, and its use as a measurable sign of how much stress a cell is under.

The GSH/GSSG redox cycle

Glutathione protects cells by handing out electrons to neutralise reactive oxygen species (unstable, cell-damaging particles). When it does this, it turns into its used-up form, GSSG. Then a helper enzyme called glutathione reductase recycles GSSG back into the fresh form, GSH, using an energy-carrying molecule called NADPH. This keeps the cell topped up with antioxidant power. This non-stop loop is the heart of how cells defend themselves.

Glutathione peroxidase and enzymatic defence

Glutathione is the fuel that a group of enzymes called glutathione peroxidases (GPx) use to break down harmful substances — like hydrogen peroxide and damaged fats — into safe ones. Through these enzymes, glutathione sits right at the centre of the cell's defence against the kind of damage studied in many cell-stress experiments.

Detoxification and conjugation

Besides acting as an antioxidant directly, glutathione also helps clear out harmful chemicals. With the help of enzymes called glutathione S-transferases, it sticks itself onto unwanted or foreign substances so the cell can

get rid of them. This clean-up job is another big area of glutathione research.

Glutathione as an oxidative-stress marker

The balance of fresh (GSH) to used-up (GSSG) glutathione shifts in a measurable way when a cell is under stress. Because of this, scientists use it as a number to track how stressed a cell is. That is one reason pure glutathione is valued as a steady, reliable reference chemical in these studies.

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ANTIOXIDANT SERIES

Melanotan II

SPECIFICATIONS

PEPTIDE CLASS	Cyclic α -MSH analogue
TARGETS	MC1R–MC5R
KEY RECEPTORS	MC1R, MC3R, MC4R
HPLC PURITY	≥99.3% (batch-verified)

What it is

Melanotan II (MT-2 for short) is a lab-made peptide (a short chain of amino acids, the building blocks of proteins). It is a copy of a natural hormone called alpha-melanocyte-stimulating hormone (α -MSH). In lab tests it flips a set of cell switches called melanocortin receptors. Scientists use it as a reference tool to study how those switches work. It is sold strictly as a research chemical for in vitro use (test-tube and lab work only). It is not a medicine, and it is not for use by people or animals, and not for tanning or cosmetic use of any kind.

MT-2 was first made at the University of Arizona in the 1980s and 1990s. Its chemical chain is joined into a ring shape, which makes it tougher and longer-lasting than the natural α -MSH hormone. There are five of these melanocortin switches, named MC1R through MC5R. In the lab MT-2 turns on four of them — MC1R, MC3R, MC4R and MC5R — but mostly leaves the fifth one (MC2R) alone. Because it activates so many at once, it is one of the broadest "switch-flippers" used in research.

That broad reach is exactly what makes it handy in the lab. The melanocortin system is involved in many things in the body, so a stable peptide that switches on several receptors gives researchers one steady reference chemical. They can line up other, more targeted chemicals against it to compare how each one behaves. These are research observations only, not effects in people.

What's been studied

■ Melanocortin receptor agonism

Melanotan II research is grouped around the cell switches it turns on: the pigment switch MC1R, the brain switches MC3R and MC4R, and its overall value as a broad reference tool.

■ Non-selective melanocortin agonism

An agonist is something that turns a receptor "on". MT-2 turns on four melanocortin switches — MC1R, MC3R, MC4R and MC5R — and mostly skips the fifth (MC2R). This is the main feature researchers study, because each switch controls a different job in the body and MT-2 hits several of them at the same time.

■ MC1R and the pigmentation pathway

MC1R is a switch found on melanocytes — the cells that make melanin, the pigment that gives skin and hair its colour. In cell and animal studies, scientists use MT-2 to turn on MC1R so they can watch how this switch controls pigment production. These are research observations only, not therapeutic or cosmetic effects in people.

■ MC3R / MC4R and central pathways

MC3R and MC4R are switches found in the brain and nerves. Researchers study them for their part in energy use and feeding (how and when animals eat). In rodent studies, MT-2 has been used to test what these

switches do for appetite, which makes it a tool for taking apart how the brain's melanocortin signals work. Again, these are research observations only, not effects in humans.

■ A reference agonist for receptor-selectivity research

Because MT-2 turns on several switches at once, scientists often use it as a "yardstick" chemical. They compare newer chemicals that target just MC3R or just MC4R against MT-2 to see how those narrower ones behave. This makes it an important lab reference in melanocortin research.

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Glossary

Plain-English definitions of the terms used throughout this handbook.

Amino acid	One of the small building blocks that join together to form peptides and proteins.
Peptide	A short chain of amino acids joined in a set order; shorter than a protein.
Lyophilised	Freeze-dried into a dry powder for stable storage; the form Velox ships.
Reconstitution	Mixing a freeze-dried powder back into a liquid so it can be measured and used.
Bacteriostatic water	Sterile water with a small amount of preservative that stops bacteria growing.
Concentration	How much peptide is dissolved in each millilitre of liquid (mg/mL).
Half-life	How long it takes for half of a substance to break down or clear away.
Purity	The proportion of a sample that is the intended compound rather than anything else.
HPLC	High-performance liquid chromatography; a lab method that measures how pure a sample is.
Mass spectrometry	A lab method that weighs a molecule precisely to confirm its identity.
CoA	Certificate of Analysis; the batch report proving a vial's purity, identity and appearance.
In vitro	"In glass" — research done in test tubes, dishes and cell cultures, not in living people.
Receptor	A "lock" on or in a cell that a molecule fits into to send a signal.
Agonist	A molecule that fits a receptor and switches it on — like a key turning a lock.
Angiogenesis	The growing of new blood vessels, often studied in tissue-repair models.
Neuropeptide	A peptide that acts on nerve cells and brain signalling.

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